Adaptive Spectral Thresholding via Kurtosis-Guided Squared Envelope Analysis for Load-Robust Bearing Fault Detection

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***Abstract*—Rolling element bearing diagnostics rely on envelope spectrum analysis to extract the periodic impulse rates produced by localized surface defects. In industrial operation, however, variable mechanical load alters the shaft slip, the structural damping and the broadband noise floor, so that fixed bandpass filters and static spectral thresholds do not transfer across load: the same absolute threshold either misses harmonics or floods the spectrum with crossings, depending on the operating point. This paper presents a training-free pure Digital Signal Processing (DSP) framework that combines Kurtosis-guided resonance demodulation with local noise-floor adaptive thresholding in the Squared Envelope Spectrum (SES). Cascade second-order-section (SOS) zero-phase IIR filtering, discrete Hilbert transform demodulation, Hann-windowed Fast Fourier Transform (FFT) and order-statistic Median–Median Absolute Deviation (Median–MAD) Constant False Alarm Rate (CFAR) thresholding allow the detector to track load-induced noise-floor shifts and shaft slip automatically. The processing chain is mathematically interpretable, requires no training data, and is sized for fixed-point embedded implementation; it is applied here to the multi-load CWRU and MFPT bearing benchmarks.**

***Index Terms*—Bearing fault diagnosis, Squared Envelope Spectrum (SES), Hilbert transform, spectral kurtosis, IIR/FIR filter design, CFAR, Median–MAD adaptive thresholding.**

I. Introduction

ROLLING element bearings carry the shaft in motors, pumps, turbines and gearboxes, and localized surface damage grows into a severe mechanical failure if it is not found early. Each time the rolling contact passes a damaged spot on an inner race, an outer race or a rolling element, it leaves a short contact disturbance that excites resonances in the bearing and housing. The useful information is the *repetition rate* of those impacts, which demodulating the resonance recovers: the principle behind envelope analysis [1], [2].

The measured vibration is not stationary, and load is the main reason: it changes the contact force and with it the strength of the defect-induced impacts, the overall vibration and the noise floor. The effective shaft speed also differs between load conditions, and local slip pulls the true impact rate away from the geometric prediction, so a fixed threshold on raw amplitude or on the spectrum cannot work at every load.

The approach here is pure DSP, not a model trained on labelled examples: every stage, from the characteristic frequencies to the filters, transforms and thresholds, can be inspected on its own, with nothing hidden in learned weights. Interpretability does not guarantee generalization, though; bearing geometry, shaft speed, sensor placement and validation data are still needed.

This paper applies two classical operations where they matter most. Kurtosis measures impulsiveness, so it finds the resonance the defect excites [3] and thereby chooses the filter band; the decision threshold is estimated *locally* in the envelope spectrum from an order statistic, which the impulsive outliers that corrupt a moving average cannot move. The resulting detector has no learned parameters and fits a fixed-point microcontroller.

The contributions of this paper are:

- a Kurtosis-guided resonance band selection procedure inside a cascade-SOS zero-phase Butterworth filter, compared against a linear-phase Kaiser FIR design on pole–zero location, magnitude response and group delay;
- a Hilbert squared-envelope demodulator with a calibrated Hann-windowed FFT, where squaring shifts the resonance to baseband by the modulation theorem and avoids the intermodulation that a square-root envelope produces;
- a Median–MAD CFAR threshold that tracks the load-dependent noise floor, with a slip-tolerant harmonic comb, tested against static thresholding and cell-averaging CFAR on multi-load data.

II. Signal Model and Theoretical DSP Framework

A. Continuous-to-Discrete Fault Signal Model

A localized bearing defect behaves like a sequence of short impacts at times $t_m = m/f_{\text{fault}} + \varepsilon_m$, where f_{fault} is the nominal repetition rate and ε_m a timing error from slip or speed fluctuation. Impact m has amplitude A_m and excites the structure with impulse response $h(t)$; background vibration and sensor noise enter as $w(t)$, so

$$x(t) = \sum_m A_m \, h(t - t_m) + w(t). \tag{1}$$

For an underdamped structural mode the causal impulse response is a decaying sinusoid,

$$h(t) = e^{-\sigma t} \cos(2\pi f_r t)\, u(t), \qquad t \geq 0, \tag{2}$$

with σ the decay rate, f_r the resonance frequency and $u(t)$ the unit step, so the response starts at the impact and decays afterwards. This repeated-impact, exponentially decaying picture is the classical localized-defect model [4].

Manuscript submitted for the ED5017 Digital Signal Processing Mini-Project. The authors are with the Department of Engineering. Roll numbers: ED24B035, ED24B052, ED24B044 and ED24B027, in author order.

Sampling with period $T_s = 1/f_s$ gives

$$x[n] = \sum_m A_m\, h[n - n_m] + w[n], \qquad n_m \approx \text{round}(f_s t_m). \quad (3)$$

Through the impact sequence $p[n] = \sum_m A_m \delta[n - n_m]$ the same signal is $x[n] = p[n] * h[n] + w[n]$, with $*$ linear convolution and a nominal impact spacing $N_p \approx f_s/f_{\text{fault}}$ samples. N_p need not be an integer, and small timing errors push the observed spectral peaks away from the ideal kinematic frequencies. Sampling must satisfy $f_s > 2f_{\max}$, and the acquisition chain has to suppress out-of-band components first or aliasing folds them into the band of interest. The records here run at 12 and 48 kHz (CWRU) and 48.828 and 97.656 kHz (MFPT), putting Nyquist at 6, 24, 24.414 and 48.828 kHz.

In the z-plane $h[n]$ gives the complex-conjugate pole pair

$$z_{1,2} = re^{\pm j\omega_r}, \qquad r = e^{-\sigma T_s} < 1, \quad (4)$$

and $|z_{1,2}| < 1$ gives stability. The model also states the central point: the fault signature A_m is a *low-frequency modulation* of a *high-frequency carrier* ω_r, which demodulation separates and direct spectral inspection does not.

B. Pre-Processing and Kurtosis-Guided Bandpass Filter Design

1) DC Offset Removal and Detrending: Sensor electronics add a constant bias, which appears as a large peak at 0 Hz whose sidelobes leak into the low-frequency bins that matter. Before any filtering, the signal $x[n]$ of length N is centered by subtracting its sample mean:

$$x_0[n] = x[n] - \mu_x, \quad \text{where} \quad \mu_x = \frac{1}{N}\sum_{m=0}^{N-1} x[m] \quad (5)$$

In the DFT sense this sets $X_0(e^{j0}) = \sum_{n=0}^{N-1} x_0[n] = 0$ and leaves every other coefficient untouched, so the filter bank sees only the vibration riding on a zero baseline.

2) Sub-Band Kurtosis Resonance Selection: Below about 1000 Hz the spectrum is crowded with shaft harmonics, structural vibration and load-dependent friction noise, which hide the bearing impulses. Each time a rolling element hits a defect it excites a high-frequency resonance of the housing, and that band carries sharp bursts while the low-frequency interference stays stationary. The resonance belongs to the structure, not to the fault, which is why one band collects many bursts at once.

The resonance depends on the machine and its mounting, so a fixed bandpass filter cannot find it reliably and the band has to be picked per record. A filter bank built on the Fast Kurtogram [3] does this without training data. The Nyquist interval $[0, f_s/2]$ is split at dyadic and half-integer levels. On a 12 kHz record the limit is 6000 Hz, so the dyadic levels give bands of 3000, 1500, 750 Hz and so on, while the half-integer levels add widths of 2000, 1000 and 545.5 Hz. The 545.5 Hz band used in Section IV comes from that set. At level k there are $M_k = \text{round}(2^k)$ bands, and band i has bandwidth and center frequency

$$\Delta f_k = \frac{f_s}{2M_k}, \quad f_{c,i} = \left(i + \frac{1}{2}\right)\Delta f_k \quad (6)$$

The search is restricted to $[1000\text{ Hz},\ f_s/2 - 150\text{ Hz}]$: the lower guard skips the crowded region below 1000 Hz and the upper guard keeps the acquisition anti-alias transition band out of the search. Any sub-band crossing either guard is scored $-\infty$, so an excluded band can never win.

For each band output $y_{k,i}[n]$, impulsiveness is measured by the fourth standardized central moment, the Kurtosis [5]:

$$K(f_{c,i}, \Delta f_k) = \frac{\frac{1}{N}\sum_{n=0}^{N-1}\left(y_{k,i}[n] - \mu_y\right)^4}{\left(\frac{1}{N}\sum_{n=0}^{N-1}\left(y_{k,i}[n] - \mu_y\right)^2\right)^2} \quad (7)$$

with μ_y the mean of the band signal. The fourth power lets one large transient outweigh many small samples, while the squared variance σ_y^4 normalizes out the band energy so a loud band does not win by being loud. Gaussian noise gives $K \approx 3$; a band of repeated impacts gives K well above 3. The band with the largest K is selected:

$$\{f_c^*, BW^*\} = \arg\max_{f_{c,i},\,\Delta f_k} K(f_{c,i}, \Delta f_k) \quad (8)$$

Its cutoffs are $f_{\text{low}} = f_c^* - BW^*/2$ and $f_{\text{high}} = f_c^* + BW^*/2$, that is $\omega_1 = 2\pi f_{\text{low}}/f_s$ and $\omega_2 = 2\pi f_{\text{high}}/f_s$.

3) Bandpass Filter Design: Kaiser FIR versus Butterworth IIR: Two filters are designed for that passband, a linear-phase FIR and a Butterworth IIR, both rejecting out-of-band motor interference and broadband noise. The FIR is Type-I, with even order M and $L = M + 1$ taps: the ideal bandpass impulse response is truncated around its symmetry center $\alpha = M/2$ and multiplied by a Kaiser window [6]:

$$h_{\text{FIR}}[n] = \left(\frac{\sin(\omega_2(n-\alpha)) - \sin(\omega_1(n-\alpha))}{\pi(n-\alpha)}\right) \times \frac{I_0\left(\beta\sqrt{1 - \left(\frac{n-\alpha}{\alpha}\right)^2}\right)}{I_0(\beta)} \quad (9)$$

for $0 \le n \le M$. $I_0(\cdot)$ is the zeroth-order modified Bessel function of the first kind, and β sets the trade between transition width and sidelobe attenuation. The taps are symmetric, $h_{\text{FIR}}[n] = h_{\text{FIR}}[M - n]$, so the phase is exactly linear and the delay is flat:

$$\tau_{g,\text{FIR}}(\omega) = -\frac{d\angle H_{\text{FIR}}(e^{j\omega})}{d\omega} = \frac{M}{2}\ \text{samples} \quad (10)$$

An FIR of this form has all its poles at $z = 0$, so it is stable whatever the coefficients do; the price is arithmetic, since every extra tap buys a steeper transition at one more operation per sample. Section IV uses $\beta = 8$, which targets roughly -80 dB of stopband attenuation.

The second design is a 4th-order Butterworth bandpass, obtained with the bilinear transform [7]. That transform maps the left half of the s-plane into the unit circle:

$$s = \frac{2}{T_s}\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right) \quad (11)$$

with $T_s = 1/f_s$. The mapping compresses frequency nonlinearly, so the digital cutoffs are pre-warped before the prototype is transformed:

$$\Omega_k = \frac{2}{T_s}\tan\left(\frac{\omega_k}{2}\right), \quad k \in \{1, 2\} \quad (12)$$

After pre-warping the −3 dB points land exactly on f_{low} and f_{high} with a maximally flat passband. Those are single-pass points: the forward–backward filter squares the magnitude response, so they appear at −6 dB in the effective zero-phase response. It runs on buffered 2.0 s blocks, so the whole block is available before the reverse pass begins.

4) Cascade SOS Realization and Zero-Phase Filtering: A 4th-order bandpass has eight poles, and in direct form a small coefficient error moves them far enough to cross the unit circle. The fix is to factor the transfer function into $K = 4$ second-order sections:

$$H_{\text{IIR}}(z) = G \prod_{k=1}^{K} H_k(z) = G \prod_{k=1}^{K} \frac{b_{0k} + b_{1k} z^{-1} + b_{2k} z^{-2}}{1 + a_{1k} z^{-1} + a_{2k} z^{-2}} \quad (13)$$

where G is the overall gain. Each section holds one conjugate pair $p_k, p_k^* = r_k e^{\pm j\theta_k}$. Stability needs the ROC $|z| > r_{\max}$ to contain the unit circle, so every pole radius must satisfy

$$r_k = \sqrt{a_{2k}} < 1, \quad \forall k \in \{1, 2, \ldots, K\} \quad (14)$$

The cascade is cheap, but its phase is not linear near the band edges, so the frequency-dependent group delay smears a short impact and lowers its peak. Forward–backward filtering removes that: the record is filtered, reversed, filtered again, and reversed back,

$$Y_{\text{BP}}(z) = H(z) H(z^{-1}) X_0(z) \quad (15)$$

On the unit circle, $H(e^{-j\omega}) = H^*(e^{j\omega})$ for real coefficients, so

$$H_{\text{zero-phase}}(e^{j\omega}) = H(e^{j\omega}) H^*(e^{j\omega}) = |H(e^{j\omega})|^2 \quad (16)$$

which is real and non-negative, so the phase and the group delay are zero. The impacts keep their time positions and shape, and the stopband attenuation doubles in decibels: −40 dB becomes −80 dB. The passband edge shifts a little inward, because the effective filter is the square of the designed one.

C. Hilbert Demodulation and Windowed Squared Envelope Spectrum

The bandpass stage leaves $y_{\text{BP}}[n]$ as short bursts of structural ringing repeating at f_{fault}. Its FFT puts the energy around the resonance ω_r, so the fault rate is nearly invisible and has to be moved to a low-frequency region where the harmonics are far apart. That operation is demodulation.

The first step is the analytic signal,

$$z[n] = y_{\text{BP}}[n] + j\hat{y}_{\text{BP}}[n], \quad (17)$$

where $\hat{y}_{\text{BP}}[n] = \mathcal{H}\{y_{\text{BP}}[n]\}$ is the Hilbert transform of the filtered signal.

The Hilbert transform alone removes nothing; it is an all-pass phase shifter, delaying positive frequencies by $\pi/2$ and advancing negative ones by $\pi/2$:

$$H_{\text{Hilbert}}(e^{j\omega}) = -j\ \text{sgn}(\omega). \quad (18)$$

The cancellation happens when the analytic signal is formed: with $Y_{\text{BP}}(e^{j\omega})$ the spectrum of the real record, $Z(e^{j\omega}) = (1 + \text{sgn}(\omega))\, Y_{\text{BP}}(e^{j\omega})$ doubles the positive-frequency half and sets the negative half to zero, and the complex $z[n]$ is what makes a well-defined envelope possible.

The usual envelope takes a square root of $y_{\text{BP}}^2 + \hat{y}_{\text{BP}}^2$. The squared envelope is used here instead:

$$e[n] = |z[n]|^2 = z[n] z^*[n] = y_{\text{BP}}^2[n] + \hat{y}_{\text{BP}}^2[n]. \quad (19)$$

Squaring in time is an auto-convolution in frequency, $E(e^{j\omega}) = \frac{1}{2\pi} Z(e^{j\omega}) * Z(e^{j\omega})$, which places a copy of the modulation spectrum at baseband and another at twice the carrier; the baseband copy is kept, so the modulation theorem recovers the fault rate without distortion. A square root is non-linear and adds intermodulation harmonics that can hide a weak fault. Squaring also favours large amplitudes: an impact peak grows much faster than the background, so the impulses stand farther above the noise than in $|z[n]|$ and no per-sample square root is needed.

The squared envelope carries a large DC term, bigger than the fault harmonics and leaking into the low-frequency bins where cage and shaft harmonics sit, so the mean is removed first:

$$e_0[n] = e[n] - \mu_e, \qquad \mu_e = \frac{1}{N} \sum_{n=0}^{N-1} e[n]. \quad (20)$$

A fault frequency rarely sits on a DFT bin centre. At 107.36 Hz the mismatch is unavoidable, and rectangular truncation spreads sidelobes into the neighbouring bins. An N-point Hann window controls them:

$$w[n] = \tfrac{1}{2}\left(1 - \cos\frac{2\pi n}{N-1}\right), \qquad n = 0, 1, \ldots, N-1. \quad (21)$$

The window brings both ends to zero, halving the coherent amplitude, and widens the mainlobe; that trade is accepted for the much lower sidelobes. The coherent gain

$$CG = \frac{1}{N} \sum_{n=0}^{N-1} w[n] = \frac{N-1}{2N} \approx \frac{1}{2} \quad (22)$$

puts the amplitude back, so the SES levels keep their physical meaning. The one-sided SES is then computed with an N_{FFT}-point radix-2 FFT:

$$\text{SES}[k] = \frac{c_k}{N \cdot CG} \left| \sum_{n=0}^{N-1} w[n]\, e_0[n]\, e^{-j\frac{2\pi}{N_{\text{FFT}}} kn} \right|, \quad (23)$$

for $k = 0, 1, \ldots, N_{\text{FFT}}/2$, where bin k corresponds to $f_k = k\, f_s / N_{\text{FFT}}$ Hz. The scale is $c_k = 2$ for the interior bins $1 \le k < N_{\text{FFT}}/2$ and $c_k = 1$ at DC ($k = 0$) and at Nyquist ($k = N_{\text{FFT}}/2$), which have no negative-frequency twin. Zero-padding a 24 000-sample frame to $N_{\text{FFT}} = 32\,768$ interpolates the DFT grid to $\Delta f \approx 0.37$ Hz, but the physical resolution is still set by the 2.0 s window (0.5 Hz).

D. Local Noise-Floor Adaptive Spectral Thresholding

Load changes the SES floor in both directions: at 0 HP the impacts are weak and a fixed threshold set for normal running sits above the fault peaks, while at 3 HP friction raises the whole floor and the same threshold is crossed by noise in dozens of bins. No single fixed value satisfies both cases. A CFAR rule [8] slides a window along the SES and

estimates the local floor from the bins around the test bin, so the comparison is always made against the floor that is actually there, with no training data.

For a test bin k, the $N_G = 4$ bins on each side are kept out of the estimate, which covers the four-bin Hann mainlobe so a fault peak cannot enter its own estimate. The $N_R = 25$ bins beyond each guard band form the reference window

$$\mathcal{W}_k = \big\{\mathrm{SES}[m] \;\big|\; m \in [k - N_G - N_R,\, k - N_G - 1] \cup [k + N_G + 1,\, k + N_G + N_R]\big\}. \tag{24}$$

Two estimators are compared on $\mathcal{W}_k$.

1) Cell-Averaging CFAR: Cell averaging takes the mean of the window, as a moving average would:

$$\hat{\mu}[k] = \frac{1}{2N_R} \sum_{x \in \mathcal{W}_k} x, \qquad T_{\mathrm{CFAR}}[k] = \alpha\, \hat{\mu}[k], \tag{25}$$

with $\alpha = 3.5$. This works while the floor changes slowly, but it fails when a harmonic enters the window: a shaft harmonic at 1× or 2× RPM or a nearby fault harmonic pulls the mean up, raising the threshold exactly where it should stay low, because a single large bin is enough to move a mean. This harmonic target masking is why the next estimator exists.

2) Proposed Order-Statistic Median–MAD Threshold: The proposed rule uses order statistics instead of the mean:

$$\begin{aligned} \hat{m}[k] &= \mathrm{median}\,(\mathcal{W}_k)\,, \\ \mathrm{MAD}[k] &= \mathrm{median}\,(|\mathcal{W}_k - \hat{m}[k]|)\,. \end{aligned} \tag{26}$$

The factor 1.4826 is the nominal scale that makes the MAD agree with the standard deviation of a normal distribution, $\hat{\sigma}_{\mathrm{local}}[k] = 1.4826\,\mathrm{MAD}[k]$, used here as a fixed scale factor rather than an unbiased Gaussian σ estimate, because the SES floor is not Gaussian. The threshold is

$$T_{\mathrm{MAD}}[k] = \hat{m}[k] + \gamma\,(1.4826\,\mathrm{MAD}[k])\,, \tag{27}$$

with $\gamma = 7.0$: a larger γ lowers the false-alarm rate and a smaller one raises it. The median has a 50% breakdown point [9], so up to half the window can be contaminated before T_{MAD} moves, which lets the threshold follow the floor from 0 HP to 3 HP.

3) Slip-Tolerant Harmonic Comb: Heavy radial load also makes the rolling elements skid by 1–2%, so the measured fault frequency is not the kinematic value and the detector searches a band, not a single bin,

$$\mathcal{B}_h = \big[h f_{\mathrm{fault}}(1 - \delta_{\mathrm{slip}}),\; h f_{\mathrm{fault}}(1 + \delta_{\mathrm{slip}})\big]\,, \tag{28}$$

for each fault frequency (f_{BPFO}, f_{BPFI} or f_{BSF}) and harmonic order $h = 1, 2, 3$. The half-width $\delta_{\mathrm{slip}} = 1.5\%$ excludes neighbouring orders while covering the slip. A harmonic is present when

$$\max_{f_k \in \mathcal{B}_h} (\mathrm{SES}[k] - T_{\mathrm{MAD}}[k]) > 0. \tag{29}$$

The exceedance rate this rule produces is not a thermal-noise false-alarm probability: on a bearing SES the eligible non-harmonic bins hold unmasked shaft harmonics and modulation sidebands, so the measured rate counts the discrete interference lines the harmonic mask did not cover.

TABLE I
IDEAL KINEMATIC FAULT FREQUENCIES CALCULATED FROM THE SPECIFIED BEARING GEOMETRIES AND SHAFT SPEEDS

Condition	Speed (rpm)	BPFO	BPFI	BSF	FTF
		(Hz)			
CWRU 0 HP	1797	107.36	162.19	70.58	11.93
CWRU 1 HP	1772	105.87	159.93	69.60	11.76
CWRU 2 HP	1750	104.56	157.94	68.74	11.62
CWRU 3 HP	1730	103.36	156.14	67.95	11.48
MFPT (25 Hz shaft)	1500	81.12	118.88	63.86	10.14

III. Experimental Setup and Datasets

A. Bearing Kinematics

Let $f_{\mathrm{rot}} = \mathrm{RPM}/60$ be the shaft frequency, n_b the number of rolling elements, d and D the element and pitch diameters and α the contact angle. With the diameter ratio $\rho = (d/D)\cos\alpha$, the ideal characteristic frequencies are

$$\begin{aligned} f_{\mathrm{BPFO}} &= \frac{n_b}{2} f_{\mathrm{rot}}\,(1 - \rho)\,, \qquad f_{\mathrm{BPFI}} = \frac{n_b}{2} f_{\mathrm{rot}}\,(1 + \rho)\,, \\ f_{\mathrm{BSF}} &= \frac{D}{2d} f_{\mathrm{rot}}\left(1 - \rho^2\right), \qquad f_{\mathrm{FTF}} = \frac{f_{\mathrm{rot}}}{2}\,(1 - \rho)\,, \end{aligned} \tag{30}$$

The CWRU drive-end bearing is an SKF 6205-2RS with nine rolling elements, a 7.94 mm element diameter, a 39.04 mm pitch diameter and a 0° contact angle. Its calculated frequencies at 1797 rpm agree with published CWRU benchmark data [10]. The MFPT bearing uses the NICE geometry documented for that dataset: eight rolling elements, 0.235 in element diameter, 1.245 in pitch diameter and 0° contact angle. Geometry goes into the calculator explicitly, so each dataset gets the right bearing.

Table I lists the CWRU frequencies at the four load settings and the MFPT frequencies at 25 Hz (1500 rpm). They are ideal targets from geometry; slip and speed variation move the real peaks, which the detector of Section II-D allows for.

B. Datasets

CWRU: the Case Western Reserve University Bearing Data Center [11] provides accelerometer recordings for normal bearings and for seeded inner-race, outer-race and rolling-element defects. A public NumPy transcription of the archive [12] keeps the time series in `.npz` files, with channel keys such as DE, FE and BA. The file name carries the shaft-speed family, the defect type, the defect size and, for many fault files, the drive-end sampling rate in a `DE12` or `DE48` suffix. The loader reads all of it from that name: RPM, fault label, defect diameter, race position and the rate suffix.

Two cases need care: normal DE files carry no rate suffix and the BA rate is never inferred, so both need a verified sampling rate rather than a silent default, and FE channels run at 12 kHz even beside `DE48` data. The load conditions are 1797 rpm (0 HP), 1772 rpm (1 HP), 1750 rpm (2 HP) and 1730 rpm (3 HP); since the dataset description quotes speeds down to about 1720 rpm, each recording takes its speed from metadata where that exists, not from the load label [10].

MFPT: the standard MATLAB recordings hold a vibration vector `gs` together with load, shaft-rate and sample-rate metadata [13]. The varied-load outer-race records cover 25, 50, 100,

TABLE II
LEAKAGE FRACTION BEYOND A WINDOW-SPECIFIC MAIN-LOBE REGION, MEASURED ON AN OFF-BIN SINUSOID AT 1467.01 Hz

Window	Leakage-energy fraction
Rectangular	0.06613550
Hann	0.00017592
Blackman	0.00000295

150, 200, 250 and 300 lb; the varied-load inner-race records cover 0, 50, 100, 150, 200, 250 and 300 lb. Baseline recordings sit at 270 lb and 97.656 kHz for 6 s, and fixed-load outer-race records use 270 lb at the same high rate. The varied-load fault records run at 48.828 kHz for 3 s.

C. Signal Preprocessing

The first step subtracts the sample mean,

$$x_0[n] = x[n] - \frac{1}{N}\sum_{n=0}^{N-1} x[n], \tag{31}$$

A constant DC offset would otherwise put energy at zero frequency. Optional peak normalization divides $x_0[n]$ by $\max|x_0[n]|$ and RMS normalization by $\sqrt{(1/N)\sum_n x_0^2[n]}$; the demeaned signal stays as the reference, and normalization serves waveform-shape comparison only, never flattening real load-dependent amplitudes in quantitative comparisons.

D. Baseline Spectra and Window Comparison

The baseline FFT runs after mean removal and optional windowing. For a window $w[n]$ the windowed transform is

$$X_w[k] = \sum_{n=0}^{N-1} x_0[n]\, w[n]\, e^{-j2\pi kn/N}. \tag{32}$$

A coherent-gain-corrected one-sided amplitude spectrum comes from dividing the magnitude by $\sum_n w[n]$ and doubling the positive-frequency bins other than DC and, for even N, the Nyquist bin. The STFT applies the same transform to successive overlapping frames, so frequency content can be followed in time.

A finite record may hold a non-integer number of cycles, and the observation window then spreads energy into neighbouring DFT bins: spectral leakage. A rectangular window has a narrow main lobe and high sidelobes; Hann and Blackman lower the sidelobes at the cost of a wider main lobe. All three are compared on the same input, first on a controlled off-bin sinusoid and then on the STFT of a record.

The controlled test uses f_s = 12 kHz, N = 2048 and a tone at 1467.01 Hz. Table II gives the fraction of FFT energy outside a window-specific main-lobe region; it is an educational comparison, not a universal standard.

E. Pipeline Parameters

Unless stated otherwise the experiments use f_s = 12 kHz, a 4th-order Butterworth bandpass in cascade SOS form with forward–backward filtering, a Hann-windowed FFT with coherent-gain calibration, N_G = 4 guard bins, N_R = 25 reference bins per side, α = 3.5 for CA-CFAR and γ = 7.0 for Median–MAD. The SES uses the next power of two above the frame length (32 768 bins for a 24 000-sample frame, $\Delta f \approx 0.37$ Hz) and the baseline spectra the full record ($\Delta f \approx 0.10$ Hz for a 121 991-sample CWRU record). Thresholds are estimated on the SES up to 650 Hz and only then cropped to the evaluation band, so the CFAR reference windows keep their right-hand cells. The static threshold is calibrated once on the first 2.0 s segment of the 0 HP outer-race reference at T_{static} = 0.00228, the largest Median–MAD level in its three BPFO harmonic windows, and then held fixed everywhere, so any change in its performance comes from the data rather than from re-tuning.

IV. Results and Discussion

A. Evaluation Metrics

Four measures are computed on the SES: spectral SNR, true detection rate, false-alarm rate and execution latency. Each is scored on every record at every load and defect size listed in Section III. Every CWRU record is split into five non-overlapping 2.0 s segments (N = 24 000 samples at 12 kHz), and each segment is processed independently; a reported number is the mean over those segments, with the sample standard deviation where stated. The MFPT records last 3 s, so each contributes one 2.0 s segment.

The data come from the CWRU 2 HP Reliance Electric rig with a dynamometer load and the 6205-2RS drive-end bearing, sampled at 12 kHz over the four load steps of Table I, with outer-race spalls of 7, 14 and 21 mil. The MFPT records add seven radial load levels from 25 to 300 lbs. The fault frequencies of Table I, f_{BPFO} = 107.36 Hz and f_{BPFI} = 162.19 Hz at no load, serve as ground truth.

1) Spectral SNR. The first measure sets the energy at the harmonics against the local noise floor,

$$\text{SNR}_{\text{SES}} = 10\log_{10}\frac{\frac{1}{M}\sum_{m=1}^{M}\max_{k\in\mathcal{B}_m}|\text{SES}[k]|^2}{\frac{1}{|\mathcal{K}_{\text{noise}}|}\sum_{k\in\mathcal{K}_{\text{noise}}}|\text{SES}[k]|^2}, \tag{33}$$

where $\mathcal{B}_m$ is the ±1.5% slip window around harmonic mf_{fault}, so the numerator is the mean of the three per-harmonic peak powers, and $\mathcal{K}_{\text{noise}}$ collects the non-harmonic bins of the evaluation band above 10 Hz. The raw baseline uses the identical estimator on $x_0[n]$: a Hann-windowed, zero-padded one-sided FFT amplitude spectrum cropped to the same band. The difference $\Delta\text{SNR} = \text{SNR}_{\text{SES}} - \text{SNR}_{\text{raw}}$ is therefore the gain the filter and envelope stages add, not an artefact of two different spectral estimators.

2) True detection rate. The second measure is the share of the targeted harmonics (1×, 2× and 3 × f_{fault}) that clear the threshold,

$$\text{TDR} = \frac{1}{M}\sum_{m=1}^{M}\mathbb{I}\left(\max_{k\in\mathcal{B}_m}(\text{SES}[k] - T[k]) > 0\right)\times 100\%, \tag{34}$$

with $\mathbb{I}(\cdot)$ the indicator function and $\mathcal{B}_m = [mf_{\text{fault}}(1 - 0.015), mf_{\text{fault}}(1 + 0.015)]$ the slip search boundary.

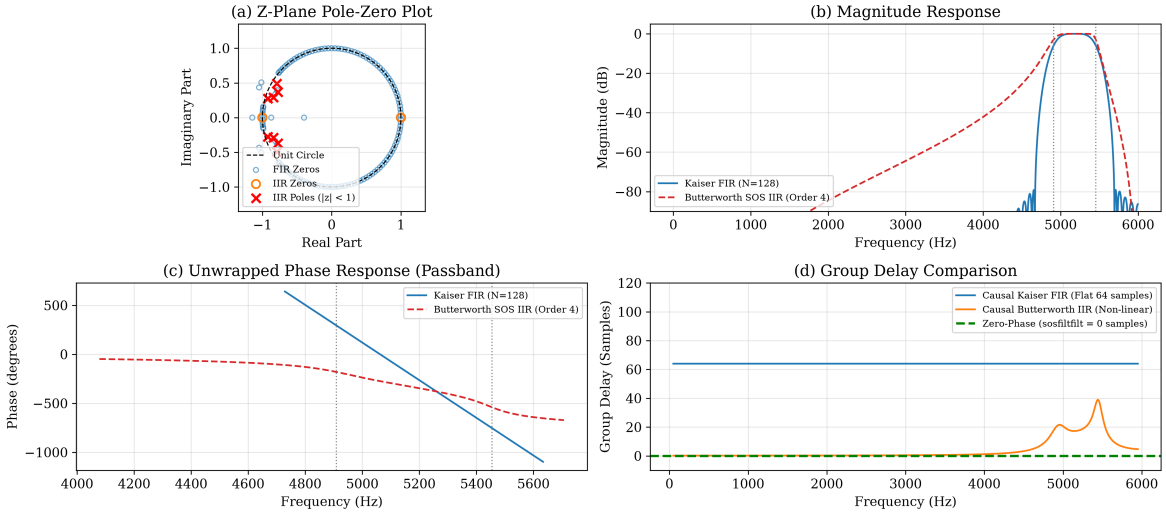


Fig. 1. Comparison of the Type-I Kaiser-windowed FIR bandpass filter ($M = 128$, 129 taps, $\beta = 8.0$) and the 4th-order Butterworth Cascade SOS IIR bandpass filter across the Kurtosis-selected passband $[4909, 5455]$ Hz: (a) complex z-plane pole–zero plot verifying that all IIR poles lie strictly inside the unit circle ($r_{\max} = 0.9626 < 1$), (b) magnitude response $|H(e^{j\omega})|$ in decibels, (c) unwrapped phase response in the passband, and (d) group delay comparison showing constant causal FIR delay (64 samples), non-linear causal IIR band-edge delay, and 0 samples net delay under zero-phase forward–backward filtering.

3) False-alarm rate. The third measure is the share of the eligible bins above 15 Hz, outside the harmonic masks, that still cross $T[k]$,

$$\text{FAR} = \frac{100\%}{|\mathcal{K}_N|} \sum_{k \in \mathcal{K}_N} \mathbb{I}(\text{SES}[k] > T[k]), \tag{35}$$

where $\mathcal{K}_N$ collects the non-harmonic bins above 15 Hz. A rate keeps records with different FFT lengths comparable.

4) Execution latency. The wall-clock time for one 2.0 s frame ($N = 24\,000$ samples at 12 kHz), taken on the same machine for every configuration.

B. Filter Architecture and Pole–Zero Stability Analysis

The Kurtogram was run on the first 2.0 s segment of the 0 HP outer-race record (7 mil defect). Over the levels spanning $[0, 6000]$ Hz it chose a band centered at $f_c^* = 5181.8$ Hz with $BW^* = 545.5$ Hz, which puts the cutoffs at 4909.1 Hz and 5454.5 Hz. Inside that band the kurtosis peaked at $K = 7.81$. The broadband input sat at $K = 7.56$.

With the band fixed, a Type-I Kaiser FIR ($M = 128$, $L = 129$ taps, $\beta = 8.0$) was compared against a 4th-order Butterworth IIR in cascade SOS form (four biquads). Fig. 1 shows the pole–zero map, the magnitude response, the phase response and the group delay.

In Fig. 1(a) the FIR has zeros along the stopband arcs and every pole at the origin, whereas the bilinear transform puts four zeros at $z = 1$ (DC) and four at $z = -1$ (Nyquist) on the IIR, with four conjugate pole pairs inside the passband arc. Its largest pole radius is $r_{\max} = 0.9626$, and $|z| > 0.9626$ contains the unit circle, so the filter is BIBO stable. Those zeros are exact nulls rather than deep attenuation, which is why DC and Nyquist are blocked completely.

Fig. 1(b) compares magnitudes: both are flat at 0 dB across $[4909, 5455]$ Hz and reject the motor region below 1000 Hz by more than 100 dB, but the 129-tap FIR rolls off faster, reaching below -80 dB within the transition and -110 dB far from the passband. The price is arithmetic: one FIR pass costs 129 multiply-accumulate (MAC) operations per sample, while the four-section SOS cascade costs $4 \times 5 = 20$ MACs per pass and its zero-phase forward–backward form $2 \times 4 \times 5 = 40$ MACs per sample, 69% fewer than the FIR.

Fig. 1(c) shows the unwrapped phase: the FIR phase is exactly linear, the signature of a symmetric Type-I design, while the IIR phase bends through the passband because its poles and zeros are not phase-matched. In Fig. 1(d) the causal FIR delays every frequency by the same 64 samples, whereas the IIR delay rises from about 20 samples at the lower cutoff to about 38 samples at 5455 Hz, where the poles approach the unit circle. Forward–backward filtering squares the magnitude, cancels the phase term and leaves zero samples of delay everywhere.

C. Time-Domain Resonance Demodulation Progression

Fig. 2 follows one segment through three stages, over 100 ms (1200 samples at 12 kHz).

In Fig. 2(a) the zero-mean record already shows the impact train: at 107.36 Hz the 100 ms window holds about eleven bursts, each an underdamped ringing of the structural resonance spaced by roughly 9.3 ms. Its kurtosis is 7.56, well above the Gaussian value of 3.0, because the impacts populate the tails of the amplitude distribution.

In Fig. 2(b) the Kurtosis-guided zero-phase SOS filter for $[4909, 5455]$ Hz removes the low-frequency shaft and motor content and leaves one resonance burst per impact on the outer race. The kurtosis moves from 7.56 to 7.81 only, since the raw record is already impulsive, but the band choice still matters for the envelope: it removes the low-frequency floor that would otherwise dominate the demodulated spectrum.

Fig. 2(c) overlays the Hilbert envelope $|z[n]|$, which follows the top of each burst and turns the 5.2 kHz carrier into a train of positive pulses spaced by $\Delta t \approx 9.31$ ms, that is $1/107.36$ Hz. The spacing of the envelope peaks, not their height, carries the

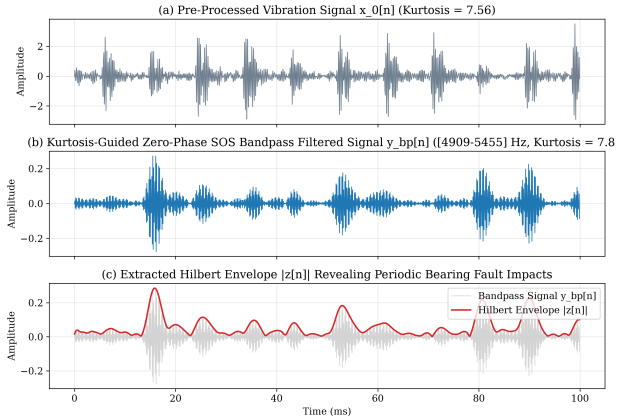


Fig. 2. Time-domain signal progression across a 100 ms segment of the 0 HP outer-race record: (a) pre-processed zero-mean vibration signal $x_0[n]$ with periodic impact bursts ($K = 7.56$), (b) Kurtosis-guided zero-phase SOS bandpass filtered output $y_{\mathrm{BP}}[n]$ across $[4909, 5455]$ Hz isolating the resonance that carries the impacts ($K = 7.81$), and (c) extracted Hilbert envelope $|z[n]|$ overlaid on $y_{\mathrm{BP}}[n]$, revealing the periodic bearing fault impulses.

fault information here, and this envelope is what the SES stage consumes.

D. Multi-Load Adaptive Thresholding

Fig. 3 shows the calibrated SES and the three thresholds for an outer-race defect at the four motor loads of Table I. The fixed level held at 0.00228 is not scale-invariant: its crossing share swings from 0% to 2% between panels and its per-segment detection rate ranges from 40 to 80% with spreads up to 55 points, so whether a harmonic clears it depends on the segment rather than on the load. Both adaptive rules follow the load-dependent floor and flag the 1×, 2× and 3× BPFO harmonics in every panel; Table III measures the difference quantitatively.

E. Quantitative Benchmark and Comparative Discussion

Table III applies the three threshold rules to the same Kurtosis-guided SES, so the detector is the only thing that changes between rows. Fig. 3 shows the spectra behind the outer-race numbers.

1) Static threshold. The fixed level is calibrated once on the 0 HP reference at $T_{\mathrm{static}} = 0.00228$, the largest Median–MAD level in that segment's three BPFO harmonic windows, and then held fixed. Even so it has no stable operating point: at 0 HP one of the five segments crosses 82% of the non-harmonic bins while the other four stay near 1%, and the detection rate moves from 0 to 100% segment by segment, giving a mean of 53.3% with a 50.6-point standard deviation; at 1 HP the mean falls to 40.0% and at 2 and 3 HP it rises to 80.0%. The spreads, up to 55 points, come from a level that sits on the edge of the SES amplitudes, so whether a harmonic clears it depends on the segment, not on the load.

2) CA-CFAR. Cell averaging follows the load-dependent floor with a false-alarm share between 5.7% and 7.9% at every load, and detects the full BPFO comb in every segment, so the mean TDR is 100% at all four loads with zero spread. Its reference window inherits energy from the fault lines themselves, which keeps the FAR from falling below about 5.7%.

3) Median–MAD. The proposed rule also returns 100% at every load while its false-alarm share stays below CA-CFAR: 6.6 against 7.0% at 0 HP, 6.9 against 7.9% at 1 HP, 4.9 against 5.7% at 2 HP and 4.7 against 6.1% at 3 HP, so the operating point is not paid for in extra crossings.

4) Full-benchmark behaviour. Across all 36 CWRU fault records (inner-race, ball and outer-race defects at 7, 14 and 21 mil, four loads each), the mean TDR is 57.8% for CA-CFAR and 56.3% for Median–MAD, against 40.9% for the fixed rule; the corresponding mean FAR is 5.0%, 4.4% and 10.7%. On the 14 MFPT variable-load fault records both adaptive rules detect the full comb (100% TDR) with FAR 2.1–2.2% (CA-CFAR) and 1.7–1.8% (Median–MAD), while the CWRU-calibrated fixed level floods the MFPT SES (88.9% and 99.6% crossings): a threshold calibrated in one dataset's units does not transfer to another. The healthy baselines show the opposite side of the same comparison: on the four normal CWRU records the fixed threshold crosses no eligible bin (0%), CA-CFAR 1.9% and Median–MAD 1.0%; on the MFPT baseline recordings CA-CFAR crosses 0.1% and Median–MAD 0.0%, while the fixed level floods 91.7% of the bins.

5) SNR gain. The filter and envelope stages add 13.4–16.4 dB of spectral SNR on the outer-race record, most at 1 HP, where the raw record sits at +5.29 dB and the SES reaches +21.71 dB. On the MFPT outer-race records the chain adds 16.1 dB, while the MFPT inner-race records already carry sharp raw harmonics so the gain is essentially zero (−0.25 dB). The detector choice does not affect this column: all three rules read the same SES.

F. Finite-Word-Length and Quantization Error Analysis

On a microcontroller or fixed-point DSP every value carries a limited number of bits, so the rounding along the chain has to be checked. The design was evaluated against IEEE 754 double precision with the coefficients in Q2.14 (one sign, one integer, fourteen fractional bits) and the signal bus in Q1.15 (one sign, fifteen fractional bits).

1) Filter structure and stability. Direct-form IIR filters are sensitive to coefficient rounding: a pole can land on the unit circle or outside it. The cascade SOS form of Section II-B keeps each conjugate pair in its own biquad, so a rounding error stays in one section. The overall gain is factored out as in Eq. (14), $G = \prod_k b_{0k}$, and each numerator is divided by its own b_{0k} so that $b_{0k} = 1$ and b_{1k}, b_{2k} stay inside $[-2, 2)$; the two sections whose $b_1/b_0 = +2$ are rescaled by $1/2$, because $+2$ lies just outside the Q2.14 range, with the factor absorbed into G. The largest pole radius is $r_{\max} = 0.96263$, and after Q2.14 rounding $r_{\max,q} = 0.96262$ is still strictly inside the unit circle, so the quantized cascade stays BIBO stable.

2) Quantization noise and SQNR. The input, every section output and the bus between sections are rounded to Q1.15; a power-of-two scale after each section keeps that bus inside $[-1, 1)$ and the overall gain is restored at the output. Rounding the high-Q denominator coefficients to Q2.14 shifts the pass-band gain slightly, so a trim evaluated on the realized cascade

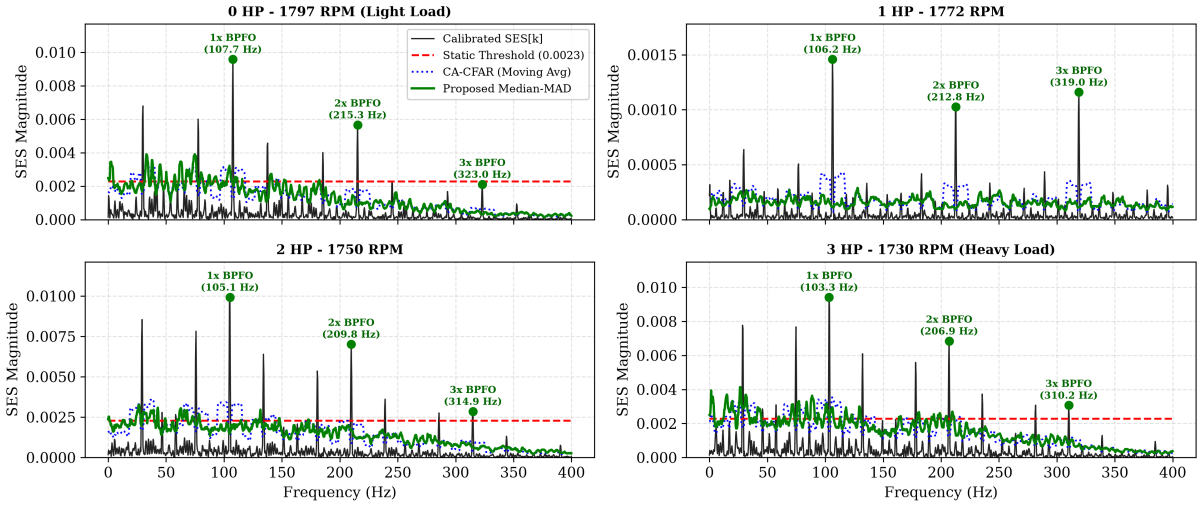


Fig. 3. Calibrated SES and comparative thresholding at four motor loads for the outer-race 7 mil record. The static threshold is calibrated once on the 0 HP reference ($T_{\text{static}} = 0.00228$); it marks two of the three BPFO harmonics in the 0 HP panel, none in the 1 HP panel and all three in the 2 and 3 HP panels. CA-CFAR (3.7–8.6%) and Median–MAD (3.6–7.4%) crossings follow the load-dependent floor and flag all three BPFO harmonics in every panel.

TABLE III

QUANTITATIVE PERFORMANCE BENCHMARKING ACROSS OPERATIONAL LOADS AND DETECTORS FOR THE OUTER-RACE 7 MIL RECORD; MEAN ± SAMPLE STANDARD DEVIATION OVER THE FIVE 2.0 S SEGMENTS OF EACH RECORD. THE STATIC RULE USES $T_{\text{STATIC}} = 0.00228$, CALIBRATED ONCE ON THE FIRST 0 HP SEGMENT AND THEN HELD FIXED; CA-CFAR USES $\alpha = 3.5$ AND THE PROPOSED RULE USES $\gamma = 7.0$. FAR IS THE PERCENTAGE OF ELIGIBLE NON-HARMONIC BINS ABOVE THE THRESHOLD.

Load	Detector	TDR (%)	FAR (%)	Raw SNR (dB)	SES SNR (dB)	ΔSNR (dB)
0 HP	Static (0.00228)	53.3 ± 50.6	17.9 ± 37.7	+8.48	+22.63	+14.15 ± 2.73
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	7.0 ± 2.1	+8.48	+22.63	+14.15 ± 2.73
	Median–MAD	100.0 ± 0.0	6.6 ± 3.3	+8.48	+22.63	+14.15 ± 2.73
1 HP	Static (0.00228)	40.0 ± 54.8	1.1 ± 1.5	+5.29	+21.71	+16.43 ± 2.70
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	7.9 ± 0.6	+5.29	+21.71	+16.43 ± 2.70
	Median–MAD	100.0 ± 0.0	6.9 ± 0.7	+5.29	+21.71	+16.43 ± 2.70
2 HP	Static (0.00228)	80.0 ± 44.7	1.7 ± 1.0	+6.41	+19.76	+13.36 ± 2.25
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	5.7 ± 1.0	+6.41	+19.76	+13.36 ± 2.25
	Median–MAD	100.0 ± 0.0	4.9 ± 0.8	+6.41	+19.76	+13.36 ± 2.25
3 HP	Static (0.00228)	80.0 ± 44.7	1.6 ± 0.9	+6.15	+19.83	+13.68 ± 1.70
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	6.1 ± 1.0	+6.15	+19.83	+13.68 ± 1.70
	Median–MAD	100.0 ± 0.0	4.7 ± 0.5	+6.15	+19.83	+13.68 ± 1.70

at the band centre is folded into G. Over the steady-state part of the filtered block, excluding the first and last 0.5 s where the zero initial state and the reverse-pass edge transient dominate, the measured signal-to-quantization-noise ratio is 70.05 dB. For $B = 16$ bits the theoretical signal-bus limit is

$$\text{SQNR}_{\text{theoretical}} \approx 6.02B + 1.76 \text{ dB} = 98.08 \text{ dB}, \qquad (36)$$

so the implementation lands about 28 dB below the ideal. The gap is coefficient-limited: Q2.14 rounds a_1 and a_2 by up to 3×10^{-5}, which for poles at radius 0.9626 already perturbs the resonance measurably. The residual $e_q[n] = y_{\text{float}}[n] - y_{Q15}[n]$ is burst-correlated rather than white (Fig. 4(b)) because the quantization error follows the impact envelope, yet the quantized envelope spectrum still overlaps the double-precision reference at the BPFO harmonics (Fig. 4(a)), so the harmonic decision is unchanged.

V. CONCLUSION AND HARDWARE FEASIBILITY

A. Summary of Diagnostic Achievements

The end-to-end run, from the raw accelerometer channel to the thresholded envelope spectrum, gives five results.

1) The Fast Kurtogram found the resonance band by itself: f_c = 5181.8 Hz with 545.5 Hz of bandwidth on the 0 HP outer-race record, with the search restricted to [1000 Hz, $f_s/2 - 150$ Hz] and no manual tuning of the filter.
2) On the outer-race 7 mil record both adaptive rules recovered the full BPFO comb in every segment at every load (100% TDR), whereas the 0 HP-calibrated fixed level averaged 40–80% with spreads up to 55 points.
3) Across all 36 CWRU fault records the adaptive rules raised the mean TDR from 40.9% (static) to 57.8% (CA-CFAR) and 56.3% (Median–MAD) and cut the mean FAR from 10.7% to 5.0% and 4.4%; on the MFPT fault records both reached 100% TDR at FAR 1.7–2.2% while the CWRU-calibrated level flooded 89–100% of the bins, and on the healthy baselines the adaptive rules stayed at 0.0–1.9% crossings.
4) Zero-phase Butterworth SOS filtering with coherent-gain calibrated Hilbert envelope analysis added 13.4–16.4 dB of spectral SNR on the CWRU outer-race record

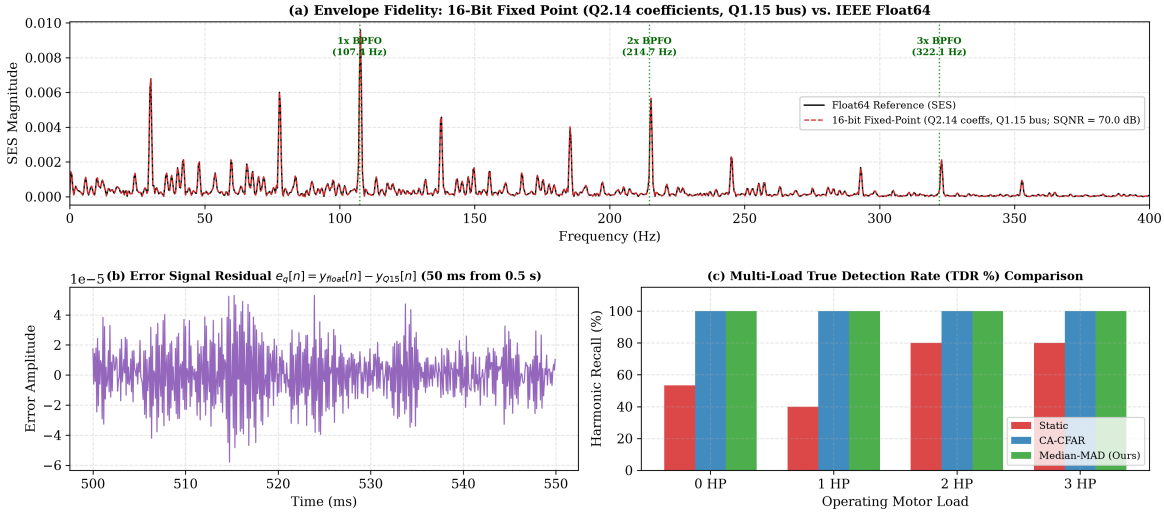


Fig. 4. Finite-word-length evaluation on the 0 HP outer-race record: (a) calibrated SES under Q2.14 coefficients and a Q1.15 signal bus against the float64 reference, (b) quantization residual $e_q[n]$ over a 50 ms window starting at 0.5 s, and (c) detector TDR across the four CWRU loads.

and 16.1 dB on the MFPT outer-race records, while the MFPT inner-race records already carried sharp raw harmonics and gained nothing (−0.25 dB).

5) With Q2.14 coefficients and a Q1.15 bus, the fixed-point cascade keeps $r_{\max,q} = 0.96262 < 1$ and reaches 70.05 dB passband SQNR, against the 98.08 dB ideal for 16 bits; the gap is set by the coefficient rounding of the high-Q denominator, not by the signal bus.

B. Embedded Hardware Feasibility and Real-Time Budget

The same pipeline was timed and re-run in fixed point, to check its fit on an edge device.

1) *Memory.* A 2.0 s block ($N = 24\,000$ samples) at 16 bits per sample needs 48 kB of SRAM and the 32 768-bin complex FFT buffer at 16 bits per component needs 128 kB, a working set of about 176 kB that fits an STM32H7 Cortex-M7 part.
2) *Execution latency.* The interpreted Python (CPython/Colab) pipeline takes 0.29–0.50 s for a 2.0 s CWRU window, so the duty cycle is

$$\eta = \frac{T_{\text{compute}}}{T_{\text{acquire}}} \approx \frac{0.29\text{–}0.50\ \text{s}}{2.0\ \text{s}} = 15\text{–}25\%. \qquad (37)$$

Even uncompiled the chain runs several times faster than real time, so the processor idles for three quarters of the timeline or more; this is a software timing, not an embedded benchmark. The MFPT records, sampled at 48.828 kHz, take 0.67–0.92 s per 2.0 s segment (34–46% duty).
3) *Fixed-point suitability.* The 70.05 dB Q1.15/Q2.14 SQNR sits 28 dB below the 98.08 dB signal-bus ideal because the high-Q denominator coefficients are rounded to 16 bits. The harmonic decision survives (Section IV-F), so integer MAC units can run the filter while the envelope FFT stage keeps a wider internal word length.

Code availability: Colab notebook.

Appendix A
CWRU Baseline Records

Figures 5–8 belong to this appendix. They come from the baseline analysis of Section III: the preprocessing chain, the static spectrum and the window comparison, all taken before any fault detection runs.

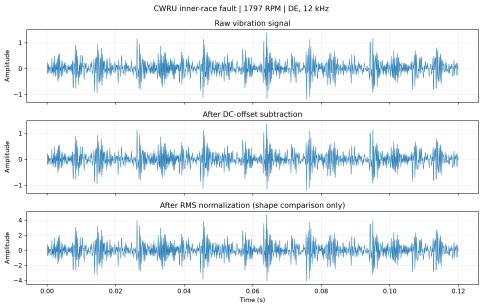


Fig. 5. CWRU inner-race fault: time-domain preprocessing (raw, DC-removed and RMS-normalized signals).

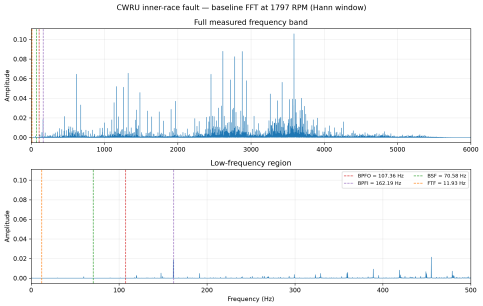


Fig. 6. CWRU inner-race fault: baseline FFT spectrum at 1797 rpm.

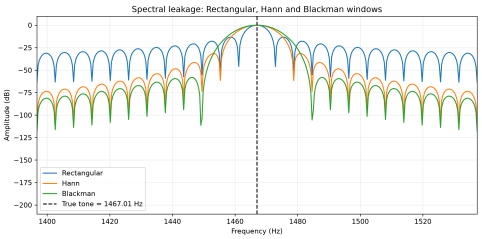


Fig. 7. Controlled off-bin single-tone comparison of the rectangular, Hann and Blackman windows.

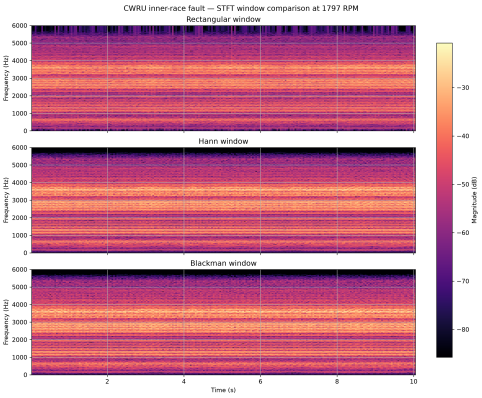


Fig. 8. CWRU inner-race fault: STFT comparison using the rectangular, Hann and Blackman windows.

Appendix B
MFPT Baseline Records

Figures 9–11 belong to this appendix. They follow the MFPT bearing through its load sweep, before and after the resonance band is selected.

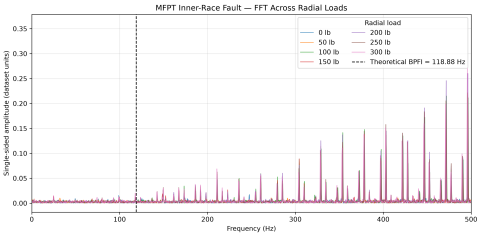


Fig. 9. MFPT inner-race fault: FFT across radial loads.

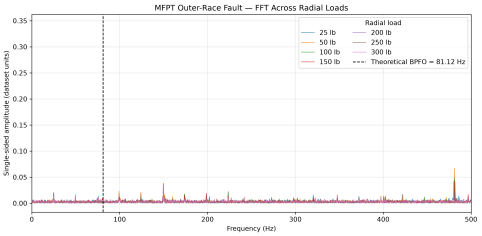


Fig. 10. MFPT outer-race fault: FFT across radial loads.

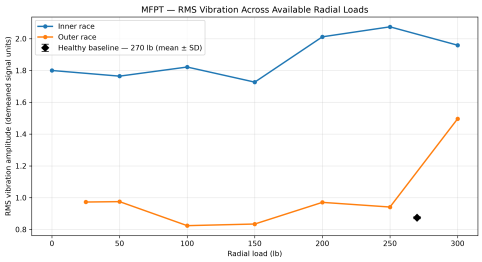


Fig. 11. MFPT RMS vibration amplitude across radial loads.

Author Contributions

Vidyashankar T curated the datasets, formulated the fault model and implemented slip-tolerant frequency tracking. **Nikunj Mayur Jethawa** designed the Kurtosis-guided band selection, implemented the FIR and cascade-SOS IIR filters and analysed poles, zeros and group delay. **Darshan L Arekal** developed the squared-envelope demodulator, calibrated the windowed SES, implemented the CA-CFAR and Median–MAD thresholding, and led the manuscript integration. **Sanjay Hariharan** ran the multi-load benchmarking and the fixed-point feasibility study, and led the manuscript integration. All authors approved the final manuscript.

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